

Algebra

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Equalities

Mastery of [Elementary Algebra](#) (a.k.a. “College Algebra”) is a prerequisite for calculus, which is in turn a prerequisite for most statistics and data science courses. Nevertheless, each year, some students are still uncomfortable with algebraic manipulations of mathematical formulas. Therefore, I include this section as a quick reference.

Theorem 0.1 (Equalities are transitive). *If $a = b$ and $b = c$, then $a = c$*

Theorem 0.2 (Substituting equivalent expressions). *If $a = b$, then for any function $f(x)$, $f(a) = f(b)$*

Inequalities

Theorem 0.3 (Adding to both sides of an inequality). *If $a < b$, then $a + c < b + c$*

Theorem 0.4 (negating both sides of an inequality). *If $a < b$, then: $-a > -b$*

Theorem 0.5 (Multiplying both sides of an inequality by a positive number). *If $a < b$ and $c > 0$, then $ca < cb$.*

Theorem 0.6 (Negation is multiplication by -1).

$$-a = (-1) * a$$

Minimum, maximum, argmin and argmax

Exercise 0.1 (Smallest value, and where it occurs).

1. Let $f(x) = (x - 2)^2 + 1$ for every $x \in \mathbb{R}$. What is the smallest value f takes, and at which x does it take that value?
2. Now let $g(x) = (x - 2)^2 + 1$ only for $x \in \{0, 1, 3, 4\}$. What is the smallest value g takes, and at which x does it take that value?

Solution

Solution 0.1.

1. The square of a real number is never negative, so $(x - 2)^2 \geq 0$ for every x , and therefore $f(x) \geq 0 + 1 = 1$. The square is 0 only when $x - 2 = 0$, that is, when $x = 2$, and there $f(2) = 1$. So the smallest value of f is 1, and $x = 2$ is the only input where f takes it.
2. Evaluate g at each of its four inputs:

x	0	1	3	4
$g(x)$	5	2	2	5

The smallest value of g is 2, and g takes it at two inputs, $x = 1$ and $x = 3$. The answer to “where?” is the set $\{1, 3\}$, not a single number.

Definition 0.1 (Minimum). Let $A \subseteq \mathbb{R}$. A number m is the **minimum** of A , written $\min A$, if $m \in A$ and $m \leq a$ for all $a \in A$.

Not every set has a minimum. The interval $(0, 1]$ has none: every element a of it has a smaller element, $a/2$, also in it. In Exercise 0.1, the smallest value of g is $\min\{5, 2, 2, 5\} = 2$.

Definition 0.2 (Maximum). Let $A \subseteq \mathbb{R}$. A number M is the **maximum** of A , written $\max A$, if $M \in A$ and $a \leq M$ for all $a \in A$.

Definition 0.3 (Argmin). Let $f : A \rightarrow \mathbb{R}$ be a **function**. The **argmin** of f over A is the set of inputs where f takes its smallest value:

$$\arg \min_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A, f(x) \leq f(a)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg \min_{x \in A} f(x)$.

The smallest value itself is $\min f(A)$, the minimum (Definition 0.1) of the **image** of f , and the argmin is where that value is attained. In Exercise 0.1, $\arg \min_{x \in \mathbb{R}} f(x) = 2$ and $\arg \min_{x \in \{0,1,3,4\}} g(x) = \{1, 3\}$. The argmin is empty when f has no smallest value, for example $f(x) = x$ on $(0, 1]$.

Definition 0.4 (Argmax). Let $f : A \rightarrow \mathbb{R}$ be a **function**. The **argmax** of f over A is the set of inputs where f takes its largest value:

$$\arg \max_{x \in A} f(x) \stackrel{\text{def}}{=} \{x \in A : \forall a \in A, f(a) \leq f(x)\}$$

When this set has exactly one element $\hat{x}$, we write $\hat{x} = \arg \max_{x \in A} f(x)$.

Global and local minimizers

Exercise 0.2 (A dip that is not the bottom). Let $f(x) = x^3 - 3x$ for $x \in \mathbb{R}$.

1. Show that $f(x) - f(1) = (x - 1)^2(x + 2)$.
2. Use part 1 to show that $f(x) \geq f(1)$ for every x in the interval $(0, 2)$.
3. Compute $f(-3)$. Is $f(1)$ the smallest value f takes on all of $\mathbb{R}$?
4. Does f take a smallest value anywhere on $\mathbb{R}$?

Solution

Solution 0.2.

1. Since $f(1) = 1 - 3 = -2$,

$$\begin{aligned} f(x) - f(1) &= x^3 - 3x + 2 \\ &= (x - 1)(x^2 + x - 2) \\ &= (x - 1)(x - 1)(x + 2) \\ &= (x - 1)^2(x + 2). \end{aligned}$$

2. For $x \in (0, 2)$, $(x - 1)^2 \geq 0$ and $x + 2 > 0$, so their product is at least 0. By part 1, $f(x) - f(1) \geq 0$, that is, $f(x) \geq f(1)$.
3. $f(-3) = -27 + 9 = -18$. Since $-18 < -2 = f(1)$, $f(1)$ is not the smallest value of f on $\mathbb{R}$.

4. No. For each $n \geq 2$, $f(-n) = -n^3 + 3n = -n(n^2 - 3) \leq -n$, so no value of f is smaller than all the others: for any x , picking $n \geq 2$ with $-n < f(x)$ gives $f(-n) < f(x)$.

Definition 0.5 (Global minimizer). Let $A \subseteq \mathbb{R}^p$ and let $f : A \rightarrow \mathbb{R}$ be a [function](#). A point $x^* \in A$ is a **global minimizer** of f over A if $f(x^*) \leq f(x)$ for all $x \in A$.

The global minimizers of f are exactly the elements of $\arg \min_{x \in A} f(x)$ (Definition 0.3). In Exercise 0.2, f has no global minimizer over $\mathbb{R}$.

Definition 0.6 (Local minimizer). Let $A \subseteq \mathbb{R}^p$ and let $f : A \rightarrow \mathbb{R}$ be a [function](#). A point $x^* \in A$ is a **local minimizer** of f if there is a number $\delta > 0$ such that $f(x^*) \leq f(x)$ for all $x \in A$ with $\|x - x^*\| < \delta$, where $\|\cdot\|$ is the [Euclidean norm](#).

For $p = 1$, $\|x - x^*\| = |x - x^*|$. Every global minimizer (Definition 0.5) is a local minimizer: take any $\delta > 0$. The converse fails. In Exercise 0.2, $x^* = 1$ is a local minimizer of f (take $\delta = 1$), but not a global one.

Infimum and supremum

Definition 0.7 (Infimum (greatest lower bound)). Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, meaning that some $t \in \mathbb{R}$ satisfies $t \leq a$ for all $a \in A$. The **infimum** of A , written $\inf A$, is the greatest real number t satisfying $t \leq a$ for all $a \in A$:

$$\inf A \stackrel{\text{def}}{=} \max\{t \in \mathbb{R} : \forall a \in A, t \leq a\}$$

If A is nonempty but not bounded below, we write $\inf A = -\infty$ by convention.

The maximum in Definition 0.7 always exists: that is the completeness (greatest-lower-bound) property of the real numbers (Rudin 1976, Definition 1.8, p. 4, and Theorem 1.19, p. 8). If the infimum belongs to A , it equals the minimum: $\inf A = \min A$.

Example 0.1 (Numerical examples of infimum).

- $\inf\{1, 2, 3\} = 1$, since 1 is the smallest element.
- $\inf(0.5, 1] = 0.5 = \min[0.5, 1]$: for intervals open below, the infimum equals the minimum of the corresponding closed-below interval, even though $0.5 \notin (0.5, 1]$. More generally, $\inf(c, b] = \min[c, b] = c$ for any $c < b$.
- $\inf\{t \geq 0 : t > 0.5\} = 0.5$, even though 0.5 itself is not in the set.

- $\inf\{-1, -2, -3, \dots\} = -\infty$, because no real number is less than or equal to every element of that set.

Definition 0.8 (Supremum (least upper bound)). Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, meaning that some $t \in \mathbb{R}$ satisfies $a \leq t$ for all $a \in A$. The **supremum** of A , written $\sup A$, is the smallest real number t satisfying $a \leq t$ for all $a \in A$:

$$\sup A \stackrel{\text{def}}{=} \min\{t \in \mathbb{R} : \forall a \in A, a \leq t\}$$

If A is nonempty but not bounded above, we write $\sup A = +\infty$ by convention.

The minimum in Definition 0.8 always exists: that is the completeness (least-upper-bound) property of the real numbers (Rudin 1976, Definition 1.8, p. 4, and Theorem 1.19, p. 8). If the supremum belongs to A , it equals the maximum: $\sup A = \max A$.

Example 0.2 (Numerical examples of supremum).

- $\sup\{1, 2, 3\} = 3$, since 3 is the largest element.
- $\sup\{t \geq 0 : t < 0.5\} = 0.5$, even though 0.5 itself is not in the set.
- $\sup\{1, 2, 3, \dots\} = +\infty$, because no real number is greater than or equal to every element of that set.

Sums

Theorem 0.7 (adding zero changes nothing).

$$a + 0 = a$$

Theorem 0.8 (Sums are symmetric).

$$a + b = b + a$$

Theorem 0.9 (Sums are associative).

When summing three or more terms, the order in which you sum them does not matter:

$$(a + b) + c = a + (b + c)$$

Products

Theorem 0.10 (Multiplying by 1 changes nothing).

$$a \times 1 = a$$

Theorem 0.11 (Products are symmetric).

$$a \times b = b \times a$$

Theorem 0.12 (Products are associative).

$$(a \times b) \times c = a \times (b \times c)$$

Division

Theorem 0.13 (Division can be written as a product). *If $b \neq 0$, then*

$$\frac{a}{b} = a \times \frac{1}{b}$$

Sums and products together

Theorem 0.14 (Multiplication is distributive).

$$a(b + c) = ab + ac$$

Exercise 0.3 (Expand a squared sum). Is $(3+4)^2$ equal to $3^2 + 4^2$? Then expand $(a+b)^2$ for any numbers a and b , using only the distributive law and the rules above.

Solution

Solution 0.3. No: $(3+4)^2 = 7^2 = 49$, while $3^2 + 4^2 = 9 + 16 = 25$. The difference, $49 - 25 = 24$, is $2 \cdot 3 \cdot 4$.

To see where that extra term comes from, write the square as a product and apply the distributive law (Theorem 0.14) twice:

$$\begin{aligned}
(a+b)^2 &= (a+b)(a+b) && \text{(definition of a square)} \\
&= (a+b)a + (a+b)b && \text{(distributive law)} \\
&= (a^2 + ba) + (ab + b^2) && \text{(distributive law, twice)} \\
&= a^2 + ab + ab + b^2 && \text{(commutative and associative laws)} \\
&= a^2 + 2ab + b^2 && \text{(collect like terms)}
\end{aligned}$$

The step “commutative and associative laws” uses Theorem 0.11 to write ba as ab , and Theorem 0.9 to drop the parentheses.

Theorem 0.15 (Square of a sum). *For any numbers a and b ,*

$$(a+b)^2 = a^2 + 2ab + b^2$$

i Proof

Proof. By Solution 0.3. □

Replacing b by $-b$ gives $(a-b)^2 = a^2 - 2ab + b^2$. Squared errors such as $(y - \hat{y})^2$ are expanded this way.

Quotients

Definition 0.9 (Quotient). For real numbers a and b with $b \neq 0$, the **quotient** of a by b is the result of dividing a by b :

$$\frac{a}{b}$$

A quotient is also called a *fraction*; a is its *numerator* and b its *denominator*. A quotient whose denominator measures time or population size is often called a *rate*; in epidemiology, rates typically have such a denominator.

cf. [https://en.wikipedia.org/wiki/Rate_\(mathematics\)](https://en.wikipedia.org/wiki/Rate_(mathematics))

Example 0.3 (A quotient). The quotient of 6 by 4 is $\frac{6}{4} = 1.5$. The quotient of 6 by 0 is undefined, because Definition 0.9 requires a nonzero denominator.

Definition 0.10 (Ratios). A **ratio** is a quotient in which the numerator and denominator are measured using the same unit scales.

cf. <https://en.wikipedia.org/wiki/Ratio>

Definition 0.11 (Proportion). In statistics, a **proportion** typically means a ratio where the numerator represents a subset of the denominator.

See https://en.wikipedia.org/wiki/Population_proportion.

See also [https://en.wikipedia.org/wiki/Proportion_\(mathematics\)](https://en.wikipedia.org/wiki/Proportion_(mathematics)) for other meanings.

Definition 0.12 (Proportional). Two functions $f(x)$ and $g(x)$ are **proportional** if their ratio $\frac{f(x)}{g(x)}$ does not depend on x . (cf. [https://en.wikipedia.org/wiki/Proportionality_\(mathematics\)](https://en.wikipedia.org/wiki/Proportionality_(mathematics)))

Additional reference for elementary algebra: https://en.wikipedia.org/wiki/Population_proportion#Mathematical_definition

Exponentials and Logarithms

In these notes, $\log\{x\}$ is the natural logarithm of $x > 0$, the logarithm with base $e \approx 2.718$, and $\exp\{x\} = e^x$ is the exponential function. Some sources write $\ln x$ for the natural logarithm and reserve $\log x$ for base 10.

Theorem 0.16 ($\exp\{\}$ and $\log\{\}$ are mutual inverses).

1. For every $a > 0$: $\exp\{\log\{a\}\} = a$.
2. For every $a \in \mathbb{R}$: $\log\{\exp\{a\}\} = a$.

Theorem 0.17 (Logarithm of a product). If $a > 0$ and $b > 0$, then

$$\log\{a \cdot b\} = \log\{a\} + \log\{b\}$$

Corollary 0.1 (Logarithm of a quotient). If $a > 0$ and $b > 0$, then

$$\log\left\{\frac{a}{b}\right\} = \log\{a\} - \log\{b\}$$

i Proof

Proof. Since $a > 0$ and $b > 0$, the quotient $\frac{a}{b}$ is positive, so Theorem 0.17 applies to the product $\frac{a}{b} \cdot b$:

$$\begin{aligned}\log\{a\} &= \log\left\{\frac{a}{b} \cdot b\right\} & (a = \frac{a}{b} \cdot b) \\ &= \log\left\{\frac{a}{b}\right\} + \log\{b\} & (\text{logarithm of a product})\end{aligned}$$

The second step applies Theorem 0.17. Subtracting $\log\{b\}$ from both sides gives $\log\{a\} - \log\{b\} = \log\{\frac{a}{b}\}$. $\square$

Theorem 0.18 (Logarithm of a power). *If $a > 0$ and $b \in \mathbb{R}$, then*

$$\log\{a^b\} = b \cdot \log\{a\}$$

Theorem 0.19 (exponential of a sum).

The exponential of a sum is equal to the product of the exponentials of the addends:

$$\exp\{a + b\} = \exp\{a\} \cdot \exp\{b\}$$

Corollary 0.2 (exponential of a difference).

The exponential of a difference is the exponential of the first term divided by the exponential of the second term:

$$\exp\{a - b\} = \frac{\exp\{a\}}{\exp\{b\}}$$

Theorem 0.20 (Power of a power). *If $a > 0$ and $b, c \in \mathbb{R}$, then*

$$a^{bc} = (a^b)^c = (a^c)^b$$

Example 0.4 (A negative base). With $a = -1$, $b = 2$, and $c = \frac{1}{2}$:

$$\begin{aligned}a^{bc} &= (-1)^{2 \cdot \frac{1}{2}} \\ &= (-1)^1 \\ &= -1\end{aligned}$$

but

$$\begin{aligned}(a^b)^c &= ((-1)^2)^{\frac{1}{2}} \\ &= 1^{\frac{1}{2}} \\ &= 1\end{aligned}$$

So $a^{bc} \neq (a^b)^c$ here, which is why Theorem 0.20 requires $a > 0$. The third expression, $(a^c)^b = ((-1)^{\frac{1}{2}})^2$, is not even a real number.

Corollary 0.3 (natural exponential of a product).

$$\exp\{ab\} = (\exp\{a\})^b = (\exp\{b\})^a$$

Exercise 0.4. For $b, c \in \mathbb{R}$, when does $b^c = bc$?

Solution

Solution 0.4. We only count a pair (b, c) when b^c is a real number, so for $b < 0$ we only consider integer c (R agrees: $(-8)^{(1/3)}$ is NaN). With that convention, $bc = b^c$ in each of the following cases:

1. $c = 1$, for every b .
2. $b = 0$ and $c > 0$, since then $b^c = 0 = bc$. ($b = 0$ and $c = 0$ fails, since $0^0 = 1 \neq 0$.)
3. $b > 0$, $c > 0$, $c \neq 1$, and $b = \exp\{\frac{\log\{c\}}{c-1}\}$. For $b > 0$, dividing both sides of $b^c = bc$ by b gives $b^{c-1} = c$, which needs $c > 0$ because $b^{c-1} > 0$; taking logarithms then gives $(c-1)\log\{b\} = \log\{c\}$. For example, $c = 2$ gives $b = 2$, and indeed $2^2 = 4 = 2 \cdot 2$.
4. $b < 0$ and c is an odd integer with $c \geq 3$, with $b = -c^{1/(c-1)}$; for example, $b = -\sqrt{3}$ and $c = 3$ give $b^c = -3\sqrt{3} = bc$.
5. $b < 0$ and c is an even integer with $c \leq -2$, with $b = -(-c)^{1/(c-1)}$; for example, $b = -2^{-1/3}$ and $c = -2$ give $b^c = 2^{2/3} = bc$.

For $b < 0$, cases 4 and 5 come from $b^{c-1} = c$ as well: when $c-1$ is even, $b^{c-1} > 0$, so c must be positive; when $c-1$ is odd, $b^{c-1} < 0$, so c must be negative.

See the red contours in Figure 2 for a visualization of the $b \geq 0$ cases.

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 1: Graph of $b * c$ and b^c

(This 3D surface is interactive; see the HTML or slide version of this page to view it.)

Figure 2: **Graph of $b^c - b * c$.** Red contour lines show where $b^c = b * c$.

Exercise 0.5. For $a \geq 0$, $b, c \in \mathbb{R}$, when does $(a^b)^c = a^{(b^c)}$?

Solution

Solution 0.5. Short answer: rarely (that's all you need to know for this course).

Long answer:

Split on whether $a > 0$ or $a = 0$, because the logarithm we use for $a > 0$ is undefined at $a = 0$.

Case $a > 0$. By Theorem 0.20, $(a^b)^c = a^{bc}$, so the question becomes when $a^{bc} = a^{(b^c)}$ (for pairs (b, c) where b^c is defined). Because $a > 0$, both sides are positive, and we can take logarithms (Theorem 0.18):

$$\begin{aligned} a^{bc} &= a^{(b^c)} \\ \log\{a^{bc}\} &= \log\{a^{(b^c)}\} \quad (\text{take logarithms of both sides}) \\ bc \cdot \log\{a\} &= b^c \cdot \log\{a\} \quad (\text{logarithm of a power}) \end{aligned} \tag{1}$$

The last line of Equation 1 holds exactly when

1. $a = 1$ (so that $\log\{a\} = 0$), or
2. $bc = b^c$ (see Exercise 0.4).

Case $a = 0$. Here we cannot take logarithms, so we work from the values of powers of 0: $0^s = 0$ for $s > 0$, $0^0 = 1$, and 0^s is undefined for $s < 0$.

- If $b < 0$, then 0^b is undefined, so $(a^b)^c$ is undefined.
- If $b > 0$, then $(0^b)^c = 0^c$ and $b^c > 0$, so $0^{(b^c)} = 0$; the two sides agree exactly when $c > 0$.
- If $b = 0$, then $(0^0)^c = 1^c = 1$; for $c > 0$, $0^{(0^c)} = 0^0 = 1$, so the two sides agree, and for $c \leq 0$ they do not.

So for $a = 0$, $(a^b)^c = a^{(b^c)}$ exactly when $b \geq 0$ and $c > 0$.

In particular, when $a = 0$, $b \geq 0$, and $c = 0$, the two sides differ:

$$\begin{aligned} (a^b)^c &= (0^b)^0 \\ &= 1 \\ a^{(b^c)} &= 0^{(b^0)} \\ &= 0^1 \\ &= 0 \end{aligned}$$

Rudin, Walter. 1976. *Principles of Mathematical Analysis*. 3rd ed. International Series in Pure and Applied Mathematics. McGraw-Hill.